Modelling of Dynamical Systems

3D printer

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Introduction

The mini-project involves the modelling of a 3D-printer in three domains: physical, discrete, and hybrid.

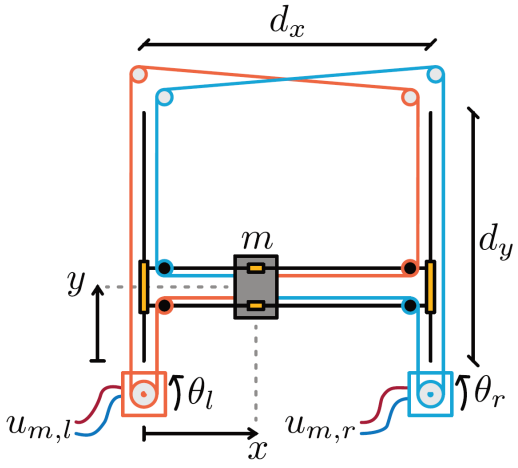


Figure 1: The schematic view of the 3D-printer.

In Figure 1 the schematic overview with the associated variable names and orientations is shown. The orientation and names of the corresponding physical quantity are respected in the derivation of the models. The mandatory variable names are given in Table 1. The formulas contain auxiliary variables which may not be given in this table.

Table 1: Variables and parameters of the 3D-printer.

Name	Symbol	Value	Unit
Motors and print head			
Motor angle left	θ_l		rad
Motor voltage left	$u_{m,l}$		V
Motor current left	$i_{m,l}$		A
Motor radius	r_m	11	mm
Motor inductance	L_m	10	mH
Motor resistance	R_m	90	Ω
Motor inertia	J_m	$8 \cdot 10^{-6}$	—
Motor constant	K_m	0.07	N m/A
Tool head position	x, y		m
Tool head mass	m	60	g
Nozzle and thermal parameters			
Nozzle voltage	u_n		V
Nozzle current	i_n		A
Nozzle capacitance	C_n		F
Nozzle thermal capacity	$C_{t,n}$		J/K
Nozzle thermal resistance	$R_{t,n}$	9	K/W
Nozzle resistance	R_n	8	Ω
Nozzle temperature	T_n	240	°C
Environmental temperature	T_e	300	K
Bearing friction			
Rolling bearing friction	μ_r	0.003	N s/m
Linear bearing friction, x-direction	$\mu_{l,x}$	0.002	N s/m
Linear bearing friction, y-direction	$\mu_{l,y}$	0.001	N s/m
Print bed and geometry			
Print bed temperature	T_b	105	°C
Print area dimension, x-direction	d_x	29	cm
Print area dimension, y-direction	d_y	29	cm

Right motor variables are analogous to those of the left motor. Empty value cells indicate variables or unspecified parameters. The unit of motor inertia is not specified in the assignment.

1 Physical Modelling

Question 1 Kinematics

The displacement of the belt by the driving motors having a radius of r_m can be expressed by the following equation

$$\Delta s = r_m \Delta\theta. \tag{1}$$

This leads to the following equations for the left and right motor being described by

$$r_m \Delta\theta_l = \Delta x + \Delta y, \qquad r_m \Delta\theta_r = \Delta x - \Delta y. \tag{2}$$

Adding and subtracting the two belt equations gives the displacements Δx and Δy, respectively:

$$\begin{aligned} 2\Delta x &= r_m \Delta\theta_l + r_m \Delta\theta_r & \quad 2\Delta y &= r_m \Delta\theta_l - r_m \Delta\theta_r \\ &= r_m(\Delta\theta_l + \Delta\theta_r), & &= r_m(\Delta\theta_l - \Delta\theta_r), \\ \Delta x &= \frac{r_m}{2}(\Delta\theta_l + \Delta\theta_r). & \Delta y &= \frac{r_m}{2}(\Delta\theta_l - \Delta\theta_r). \end{aligned} \tag{3}$$

When the reference is taken to be zero (i.e., $x_0, y_0, \theta_{l,0}, \theta_{r,0} = 0$), the prefix Δ can be discarded. The earlier derived equations simplify to

$$x = \frac{r_m}{2}\left(\theta_l + \theta_r\right), \qquad y = \frac{r_m}{2}\left(\theta_l - \theta_r\right). \tag{4}$$

The velocities can be found by differentiation

$$\dot{x} = \frac{r_m}{2}\left(\dot{\theta}_l + \dot{\theta}_r\right), \qquad \dot{y} = \frac{r_m}{2}\left(\dot{\theta}_l - \dot{\theta}_r\right). \tag{5}$$

The rotational speed is given by the derivative of the angle (i.e., $\dot{\theta} = \omega$). Therefore, the velocities can alternatively be denoted by

$$\dot{x} = \frac{r_m}{2}\left(\omega_l + \omega_r\right), \qquad \dot{y} = \frac{r_m}{2}\left(\omega_l - \omega_r\right). \tag{6}$$

For completeness, the accelerations are the second derivatives

$$\ddot{x} = \frac{r_m}{2}\left(\dot{\omega}_l + \dot{\omega}_r\right), \qquad \ddot{y} = \frac{r_m}{2}\left(\dot{\omega}_l - \dot{\omega}_r\right). \tag{7}$$

Question 2 Dynamics

The model of the DC motor coupled to a disk with an inertia is shown in Figure 2.

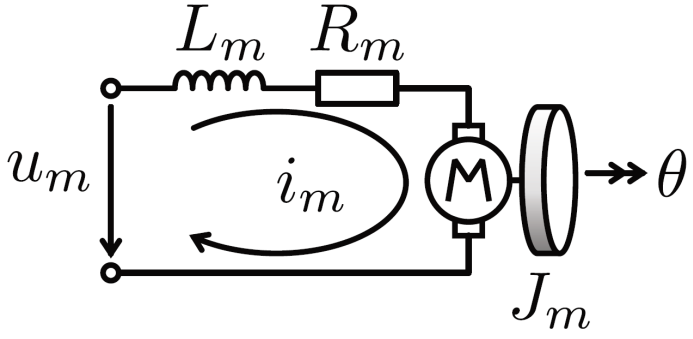


Figure 2: The proposed DC motor modelled as an LR circuit.

2.1 Electrical dynamics

The torque of the motor is related to the current flowing through the motor $\tau_m = K_m i$. Applying Kirchhoff's voltage law to the schematic of Figure 2 leads to[1]

$$u_m = u_{L_m} + u_{R_m} + u_{\text{EMF}}, \tag{8}$$

$$u_m = L_m \frac{di_m}{dt} + R_m i_m + K_m \dot{\theta}, \tag{9}$$

$$L_m \frac{di_m}{dt} = -R_m i_m - K_m \omega + u_m. \tag{10}$$

2.2 Mechanical dynamics

The printer head is modelled as a mass. The belts driven by the motors exert forces on the mass denoted by F_x and F_y in Figure 3. Friction due to the linear bearings causes an opposite force acting on the mass. This effect is modelled by the $\mu_{l,x}\dot{x}$ and $\mu_{l,y}\dot{y}$ vectors.

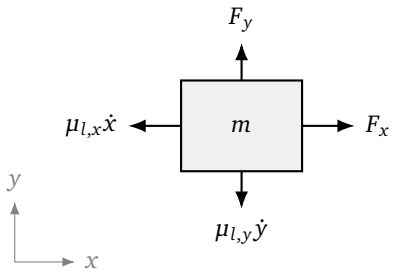


Figure 3: In-plane forces acting on the print head. Friction forces are shown for positive velocities.

Euler-Lagrange The Euler-Lagrange method can be used in order to efficiently derive the mechanical model for the system. The generalized coordinates and velocities are defined as

$$q = \begin{bmatrix} \theta_l \\ \theta_r \end{bmatrix}, \qquad \dot{q} = \begin{bmatrix} \omega_l \\ \omega_r \end{bmatrix}. \tag{11}$$

These states directly determine the x and y velocity of the printer head via Equation (6).

The kinetic energy of the motor and printer head can be combined to determine the K function of Euler-Lagrange

$$K = \underbrace{\frac{1}{2} J_m \left(\omega_l^2 + \omega_r^2\right)}_{\text{Motor}} + \underbrace{\frac{1}{2} m \left(\dot{x}^2 + \dot{y}^2\right)}_{\text{Printer head}}. \tag{12}$$

Substitution of the found kinematics with Equation (6) transforms the equation to

$$K = \frac{1}{2} J_m \left(\omega_l^2 + \omega_r^2\right) + \frac{1}{2} m \left(\left(\frac{r_m}{2}\left(\omega_l + \omega_r\right)\right)^2 + \left(\frac{r_m}{2}\left(\omega_l - \omega_r\right)\right)^2\right), \tag{13}$$

$$L = K = \frac{1}{2} \underbrace{\left(J_m + \frac{m r_m^2}{2}\right)}_{J_{\text{eff}}} \left(\omega_l^2 + \omega_r^2\right). \tag{14}$$

[1] The derivative is normally denoted by a dot (e.g., $\dot{\theta}$) in this report except when the differential quantity is current (i.e., i). In that case, the explicit notation is used $\left(\text{i.e., } \frac{di}{dt}\right)$ in order to avoid confusion due to the double dot (i.e., $\dot{i}$).

The Lagrangian function is defined by $L = K - P$. As there is no potential energy stored in the model (i.e, the belts are considered non-elastic and constant height) the function $P = 0$. Therefore, $L = K$.

The dissipation function consists of the linear and pulley bearings (i.e., $D = D_{\text{lin}} + D_{\text{pull}}$). The dissipation function for the linear bearings can be modeled and rewritten as

$$D_{\text{lin}} = \frac{1}{2}\mu_{l,x}\dot{x}^2 + \frac{1}{2}\mu_{l,y}\dot{y}^2, \tag{15}$$

$$D_{\text{lin}} = \frac{r_m^2}{8}\left[\mu_{l,x}\left(\omega_l + \omega_r\right)^2 + \mu_{l,y}\left(\omega_l - \omega_r\right)^2\right]. \tag{16}$$

It can be observed from Figure 1 that each motor drives a belt connected to the mass along two fixed pulleys and two pulleys attached to the printer head rail. For the fixed pulley, its rotational speed is equal to $\omega_p = \frac{r_m}{r_p}\omega_l$. Therefore, the dissipation function becomes

$$D_{\text{pul,fixed}} = 2\frac{1}{2}\mu_r\left(r_p\omega_p\right)^2,$$

$$D_{\text{pul,fixed}} = 2\frac{1}{2}\mu_r\left(r_m\omega_l\right)^2.$$

The centers of the four pulleys attached to the printer head rail move in the y-direction. The dissipation term for a pulley with a relative tangential speed v_p is stated as $D_p = \frac{1}{2}\mu_r v_p^2$. The relative tangential speed is equal to $v_p = \dot{x}$. Therefore,

$$D_{\text{pul,rail}} = 4\frac{1}{2}\mu_r\dot{x}^2. \tag{17}$$

Adding the dissipation terms together (i.e, $D = D_{\text{lin}} + D_{\text{pul,fixed,l}} + D_{\text{pul,fixed,r}} + D_{\text{pul,rail}}$) and substitution of Equation (6) leads to

$$D(\omega_l, \omega_r) = \frac{r_m^2}{8}\left[\mu_{l,x}\left(\omega_l + \omega_r\right)^2 + \mu_{l,y}\left(\omega_l - \omega_r\right)^2\right] + \mu_r r_m^2\left(\omega_l^2 + \omega_r^2\right) + \frac{1}{2}\mu_r r_m^2\left(\omega_l + \omega_r\right)^2. \tag{18}$$

All elements of the *Lagrange's equation of motion* are present. From this equation, the complete mechanical model can be derived

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = -\frac{\partial D}{\partial \dot{q}} + \underbrace{\begin{bmatrix} K_m i_{m,l} \\ K_m i_{m,r} \end{bmatrix}}_{\tau_m}. \tag{19}$$

From Equation (14) it follows that

$$\frac{\partial L}{\partial q} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \frac{d}{dt}\frac{\partial L}{\partial \dot{q}} = J_{\text{eff}}\begin{bmatrix} \dot{\omega}_l \\ \dot{\omega}_r \end{bmatrix}. \tag{20}$$

From Equation (18) the dissipation element is determined by

$$\frac{\partial D}{\partial \omega_l} = \underbrace{\frac{r_m^2}{4}\left[\mu_{l,x} + \mu_{l,y} + 12\mu_r\right]}_{d_s}\omega_l + \underbrace{\frac{r_m^2}{4}\left[\mu_{l,x} - \mu_{l,y} + 4\mu_r\right]}_{d_c}\omega_r \tag{21}$$

$$\frac{\partial D}{\partial \omega_r} = \underbrace{\frac{r_m^2}{4}\left[\mu_{l,x} - \mu_{l,y} + 4\mu_r\right]}_{d_c}\omega_l + \underbrace{\frac{r_m^2}{4}\left[\mu_{l,x} + \mu_{l,y} + 12\mu_r\right]}_{d_s}\omega_r. \tag{22}$$

Which can be written compactly as

$$\frac{\partial D}{\partial \dot{q}} = \begin{bmatrix} d_s\omega_l + d_c\omega_r \\ d_c\omega_l + d_s\omega_r \end{bmatrix}. \tag{23}$$

Substitution into Equation (19) results in

$$J_{\text{eff}}\dot{\omega}_l = K_m i_{m,l} - d_s\omega_l - d_c\omega_r, \tag{24}$$
$$J_{\text{eff}}\dot{\omega}_r = K_m i_{m,r} - d_c\omega_l - d_s\omega_r. \tag{25}$$

2.3 State-space model

The states are chosen as

$$\mathbf{x} = \begin{bmatrix} \theta_l, & \theta_r, & \omega_l, & \omega_r, & i_{m,l}, & i_{m,r} \end{bmatrix}^\mathsf{T}. \tag{26}$$

The input vector is defined by

$$\mathbf{u} = \begin{bmatrix} u_{m,l}, & u_{m,r} \end{bmatrix}^\mathsf{T}. \tag{27}$$

The position of the printer head is selected as the output vector of the system

$$\mathbf{y} = \begin{bmatrix} x, & y \end{bmatrix}^\mathsf{T}. \tag{28}$$

This yields the following state-space representation

$$\dot{\mathbf{x}} = \underbrace{\begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{d_s}{J_{\text{eff}}} & -\frac{d_c}{J_{\text{eff}}} & \frac{K_{m,l}}{J_{\text{eff}}} & 0 \\ 0 & 0 & -\frac{d_c}{J_{\text{eff}}} & -\frac{d_s}{J_{\text{eff}}} & 0 & \frac{K_{m,r}}{J_{\text{eff}}} \\ 0 & 0 & -\frac{K_{m,l}}{L_{m,l}} & 0 & -\frac{R_{m,l}}{L_{m,l}} & 0 \\ 0 & 0 & 0 & -\frac{K_{m,r}}{L_{m,r}} & 0 & -\frac{R_{m,r}}{L_{m,r}} \end{bmatrix}}_{\mathbf{A}} \mathbf{x} + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{L_{m,l}} & 0 \\ 0 & \frac{1}{L_{m,r}} \end{bmatrix}}_{\mathbf{B}} \mathbf{u} \tag{29}$$

$$\mathbf{y} = \underbrace{\begin{bmatrix} \frac{r_m}{2} & \frac{r_m}{2} & 0 & 0 & 0 & 0 \\ \frac{r_m}{2} & -\frac{r_m}{2} & 0 & 0 & 0 & 0 \end{bmatrix}}_{\mathbf{C}} \mathbf{x}.$$

This model assumes that the

- belts are non-elastic and do not slip,
- motor parameters are equal for the left and right motor,
- mass of the printer head rail, pulleys, and inertia of the pulleys can be neglected,
- parameters $\mu_{l,x}, \mu_{l,y}$ are total linear damping coefficients per direction,
- parameter μ_r denotes the tangential damping per pulley.

Question 3 Decouple the x and y dynamics

3.1 Decoupling equations

One can rewrite Equation (6) by choosing new velocity variables $\dot{x} = v_x, \dot{y} = v_y$. This results in

$$v_x = \frac{r_m}{2}\left(\omega_l + \omega_r\right), \qquad v_y = \frac{r_m}{2}\left(\omega_l - \omega_r\right). \tag{30}$$

When Equations (24) and (25) are added to each other, each side is multiplied with $\frac{r_m}{2}$ and Equation (30) are substituted this leads to

$$J_{\text{eff}}\dot{v}_x = \frac{r_m K_m}{2}\underbrace{\left(i_{m,l} + i_{m,r}\right)}_{i_x} - \left(d_s + d_c\right)v_x. \tag{31}$$

Similarly, for the movement along the y-direction, one can write

$$J_{\text{eff}}\dot{v}_y = \frac{r_m K_m}{2}\underbrace{\left(i_{m,l} - i_{m,r}\right)}_{i_y} - \left(d_s - d_c\right)v_y. \tag{32}$$

The electrical equations can be transformed similarly by adding Equation (10) for the left and subtraction for the right motor

$$L_{m,l}\frac{di_x}{dt} = -R_m i_x - \frac{2K_m}{r_m}v_x + \underbrace{\left(u_{m,l} + u_{m,r}\right)}_{u_x}, \tag{33}$$

$$L_{m,r}\frac{di_y}{dt} = -R_m i_y - \frac{2K_m}{r_m}v_y + \underbrace{\left(u_{m,l} - u_{m,r}\right)}_{u_y}. \tag{34}$$

3.2 Transformed state-space model

In order to decouple the x and y coordinates two state-space models are presented. The two state vectors are chosen as

$$\mathbf{x}_x = \left[x,\ v_x,\ i_x\right]^\mathsf{T}, \qquad \mathbf{x}_y = \left[y,\ v_y,\ i_y\right]^\mathsf{T}. \tag{35}$$

The inputs of the two models are represented by u_x, u_y. The position of the printer head is selected as x, y. Resulting in the transformed state-space representation

x-direction

$$\dot{\mathbf{x}}_x = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -\frac{d_s+d_c}{J_{\text{eff}}} & \frac{r_m K_m}{2J_{\text{eff}}} \\ 0 & -\frac{2K_m}{r_m L_m} & -\frac{R_m}{L_m} \end{bmatrix}\mathbf{x}_x + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L_m} \end{bmatrix}u_x,$$

$$x = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}\mathbf{x}_x. \tag{36}$$

y-direction

$$\dot{\mathbf{x}}_y = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -\frac{d_s-d_c}{J_{\text{eff}}} & \frac{r_m K_m}{2J_{\text{eff}}} \\ 0 & -\frac{2K_m}{r_m L_m} & -\frac{R_m}{L_m} \end{bmatrix}\mathbf{x}_y + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L_m} \end{bmatrix}u_y,$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}\mathbf{x}_y. \tag{37}$$

Question 4 Drawing shapes

The corresponding motor voltages need to be determined in order to draw an ellipse with the 3D-printer. The x, and $y-$coordinates can be stated as

$$x_{\text{ref}} = x_c + a\cos(\Omega t), \qquad y_{\text{ref}} = y_c + b\sin(\Omega t). \tag{38}$$

Whereas $a, b \neq 0\,\text{m}$, $a \neq b$ and $\Omega = \frac{2\pi}{T}$. A circle is drawn when $a = b$.

Using the decoupled state-space matrices (i.e., Equations (36) and (37)), the motor voltages can be derived

$$\begin{aligned} J_{\text{eff}}\ddot{x}_{\text{ref}} &= \frac{r_m K_m}{2} i_x - (d_s + d_c)\,\dot{x}_{\text{ref}}, \\ i_{x,\text{ref}} &= \frac{2}{r_m K_m}\left[J_{\text{eff}}\ddot{x}_{\text{ref}} + (d_s + d_c)\,\dot{x}_{\text{ref}}\right], \\ u_x &= L_m \frac{di_{x,\text{ref}}}{dt} + R_m i_{x,\text{ref}} + \frac{2K_m}{r_m}\dot{x}_{\text{ref}}. \end{aligned} \qquad \begin{aligned} J_{\text{eff}}\ddot{y}_{\text{ref}} &= \frac{r_m K_m}{2} i_y - (d_s - d_c)\,\dot{y}_{\text{ref}}, \\ i_{y,\text{ref}} &= \frac{2}{r_m K_m}\left[J_{\text{eff}}\ddot{y}_{\text{ref}} + (d_s - d_c)\,\dot{y}_{\text{ref}}\right], \\ u_y &= L_m \frac{di_{y,\text{ref}}}{dt} + R_m i_{y,\text{ref}} + \frac{2K_m}{r_m}\dot{y}_{\text{ref}}. \end{aligned} \tag{39}$$

When $a = 0.1\,\text{m}$, $b = 0.07\,\text{m}$, $x_{\text{ref}} = \frac{dx}{2} = 0.145\,\text{m}$, $y_{\text{ref}} = \frac{dy}{2} = 0.145\,\text{m}$ and $T = 10$ the model returns the following plot given in Figure 4.

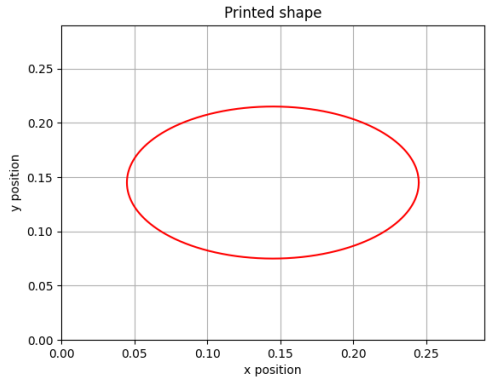


Figure 4: The ellipse drawn for $a = 0.1$ and $b = 0.07$.

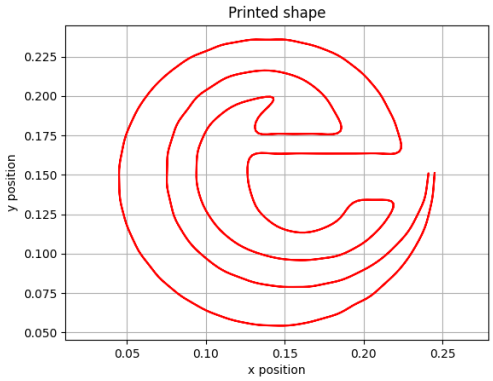


Figure 5: The Efteling logo. Drawn with $k = 250$ in Equation (40).

From Equation (38) it can be observed that using geometric functions an arbitrary shape can be drawn. Using the complex Fourier series one can convert the contour of an image in a mathematical relation. The complex Fourier series and its relation to the printer head position is given by

$$z(t) = \sum_k c_k e^{ik\Omega t}, \qquad x_{\text{ref}}(t) = \text{Re}\{z(t)\}, \qquad y_{\text{ref}}(t) = \text{Im}\{z(t)\}. \tag{40}$$

The website [1] is used to obtain the coefficients c_k. When the coefficients from [2] are used, the 3D-printer draws Figure 5.

Question 5 Model the Nozzle

The lumped model of the nozzle is shown in Figure 6.

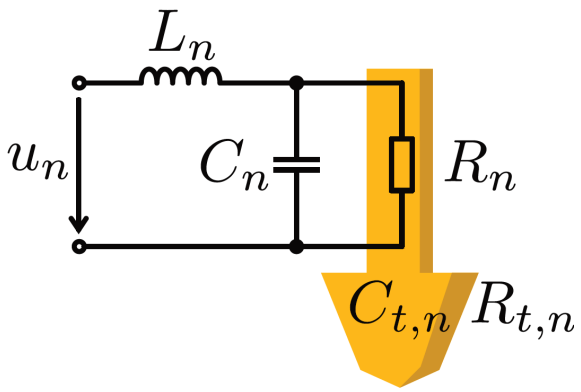


Figure 6: The lumped model of the nozzle in the electrical and thermal domain.

Assuming that all energy dissipated in R_n is converted to heat and defining the voltage across R_n as v_n, one can write

$$P_{\text{heat}} = v_n i_R = \frac{v_n^2}{R_n}. \tag{41}$$

The thermal capacity determines the amount of energy needed to increase the temperature by

$$C_{t,n}\dot{T}_n = P_{\text{heat}} - \frac{T_n - T_e}{R_{t,n}}. \tag{42}$$

Substitution of Equation (41) and solving for $\dot{T}_n$ leads to

$$\dot{T}_n = \frac{1}{C_{t,n}}\left[\frac{v_n^2}{R_n} - \frac{T_n - T_e}{R_{t,n}}\right]. \tag{43}$$

Applying Kirchhoff's voltage and current laws leads to

$$L_n \frac{di_L}{dt} = u_n - v_n, \tag{44}$$

$$C_n \dot{v}_n = i_L - \frac{v_n}{R_n}. \tag{45}$$

A suitable choice for the state vector is

$$\mathbf{x}_n = [T_n,\ i_L,\ v_n]^\mathsf{T}. \tag{46}$$

Transforming into the requested format (i.e, $\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u})$) yields

$$\frac{d}{dt}\begin{bmatrix} T_n \\ i_L \\ v_n \end{bmatrix} = \begin{bmatrix} \frac{1}{C_{t,n}}\left(\frac{v_n^2}{R_n} - \frac{T_n - T_e}{R_{t,n}}\right) \\ \frac{u_n - v_n}{L_n} \\ \frac{i_L - \frac{v_n}{R_n}}{C_n} \end{bmatrix}. \tag{47}$$

Question 6 Printing a circle of filament (PDE)

To aid in the derivation, Figure 7 is created to define the conventions.

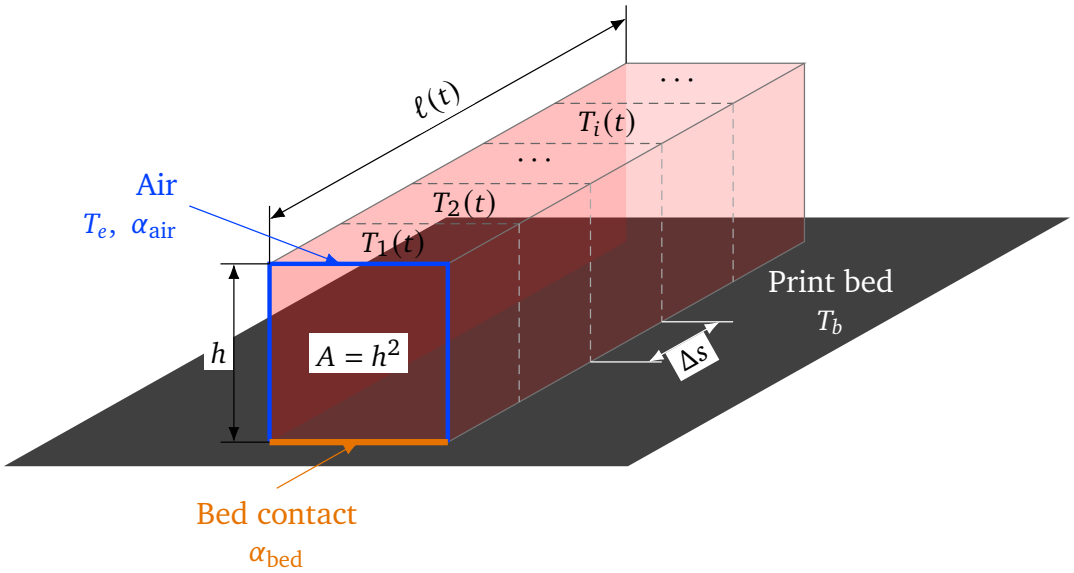


Figure 7: Spatial discretization of the deposited filament. Blue boundaries exchange heat with the air, while the orange boundary exchanges heat with the print bed.

For a distance along the printed path, it can be defined

$$0 \leq s \leq \ell(t), \qquad \dot{\ell}(t) = v_{\text{print}}(t). \tag{48}$$

The printing speed is given by

$$v_{\text{print}} = \sqrt{\dot{x}^2 + \dot{y}^2}. \tag{49}$$

The functions used for x and y are given in Equation (38).

The cross-sectional area between two pieces of filament is given by $A = h^2$. The boundary between the air and the filament are given by $p_{\text{air}} = 3h$. The print bed and the filament are in contact $p_{\text{bed}} = h$.

The Partial Differential Equation (PDE) for the heat transfer can be modelled as

$$\underbrace{\rho C_p A \frac{\partial T(s,t)}{\partial t}}_{\text{Heat storage}} = \underbrace{\lambda A \frac{\partial^2 T(s,t)}{\partial s^2}}_{\text{Conduction}} - \underbrace{\alpha_{\text{air}} 3h \left(T(s,t) - T_e\right)}_{\text{Convection}} + \underbrace{\alpha_{\text{bed}} h \left(T_b - T(s,t)\right)}_{\text{Heated bed}}. \tag{50}$$

The initial value condition can be found by defining $t_d(s)$ which indicates the time when the filament at position s is printed. At that time and position, the temperature of the filament is equal to the nozzle temperature $T(s, t_d(s)) = T_n$.

The printing temperature of the nozzle is $T_n = 240\,^\circ\text{C}$. A suitable choice for filament is ABS as this temperature falls in the generic print temperature [3]. The material parameters of [3] are used in the PDE evaluation. The effective heat contact coefficient α_{bed} is selected as $50\,\text{W}/(\text{m}^2\,\text{K})$ which is found to be adequate to illustrate the progression of the PDE versus time.

Table 1. Material input parameters [26].

Material	Density ρ (kg/m^3)	Specific Heat C_P (J/kg·K)	Thermal Conductivity λ (W/m·K)	Length L (mm)	Thickness ϕ (mm)	Height h (mm)
ABS	1050	2080	0.177	50	~0.7	~0.2

h_{ABS} (ABS convective heat transfer coefficient considered in the simulations) = 250 (W/m^2·K).

Figure 8: The ABS material properties as presented in [3]. These values are treated as generic parameters for ABS filament and discrepancies between different colors may be present.

In order to evaluate Equation (50) one can use the Method of Lines (MOL) in order to numerically compute the solution by Python. The derivative is approximated by means of the *Forward Euler Method* (i.e, $\frac{du_i}{dt} \approx \frac{u_i^{n+1}-u_i^n}{\Delta t}$). Substitution in Equation (50) leads to

$$\frac{dT_i}{dt} \approx \underbrace{\frac{\lambda}{\rho C_p}}_{a_{\text{th}}} \frac{T_{i+1} - 2T_i + T_{i-1}}{\Delta s^2} - \underbrace{\frac{3\alpha_{\text{air}}}{\rho C_p h}}_{\beta_{\text{air}}} (T_i - T_e) + \underbrace{\frac{\alpha_{\text{bed}}}{\rho C_p h}}_{\beta_{\text{bed}}} (T_b - T_i)\,. \tag{51}$$

The PDE is numerically evaluated using Python. The x and y-coordinates as a function of time and temperature of the filament are presented in the animation of Figure 9.

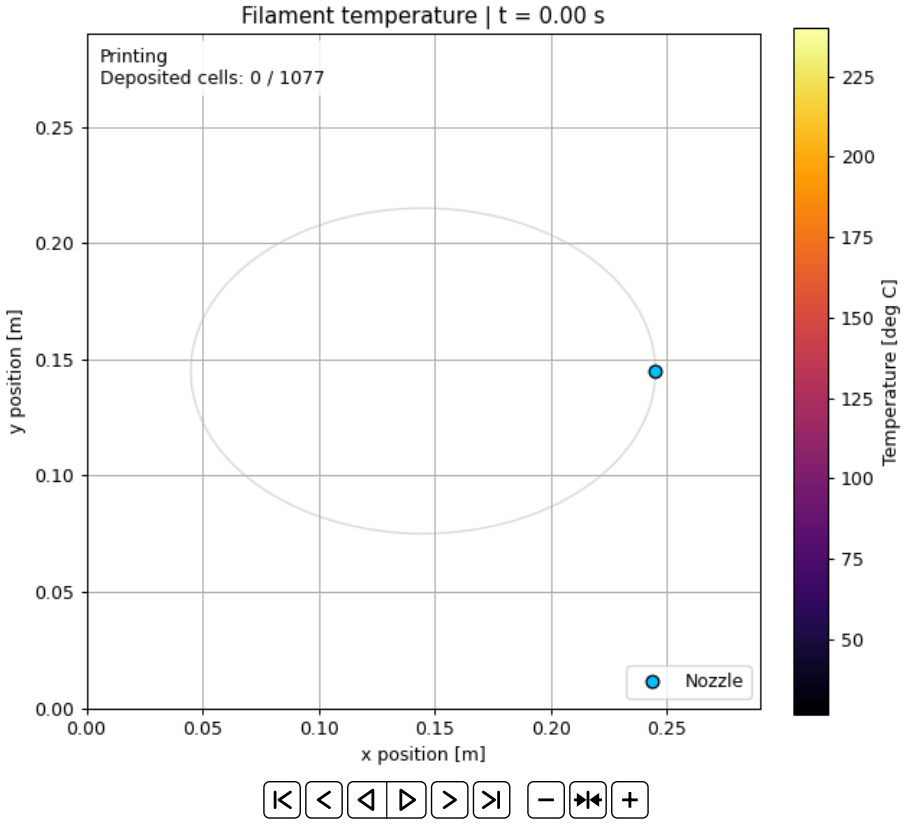


Figure 9: Printing and cooling of the filament. Open in Acrobat Reader in order to start the animation.

As time progresses, the temperature of the filament reaches a stationary value of

$$T_\infty = \frac{3\alpha_{\text{air}} T_e + \alpha_{\text{bed}} T_b}{3\alpha_{\text{air}} + \alpha_{\text{bed}}} = \frac{3 \cdot 25 \cdot 300 + 50 \cdot 378.15}{3 \cdot 25 + 50} = 331.26\,\text{K} = 58.11\,^\circ\text{C}. \tag{52}$$

In Figure 10 the temperature versus time graph of the first and last segment are presented. When the ellipse closes, the first and last segment come into contact. The last element initially heats up the first segment after which the temperature of both segments approach the stationary value calculated in Equation (52).

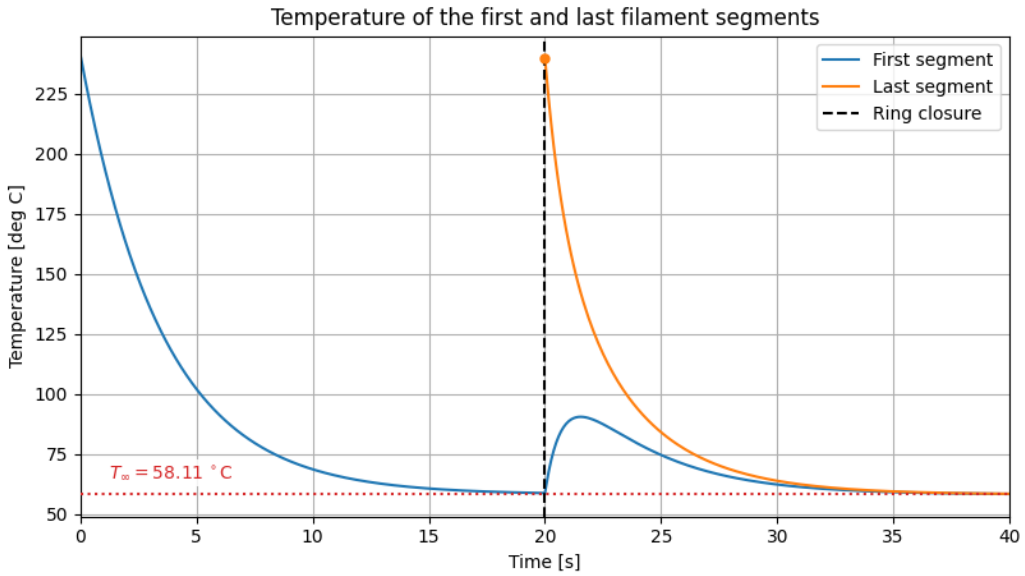


Figure 10: Temperature of the first and last segment of the printed ellipse. When the ring is closed, the first element heats up due to the thermal conductivity of the filament.

2 Finite State Machine

Question 1 Finite State Machines — Filament replacement

The Finite State Machine (FSM) is presented in Figure 11. The definitions of the states are given in Table 2. The meaning of the events are summarized in Table 5.

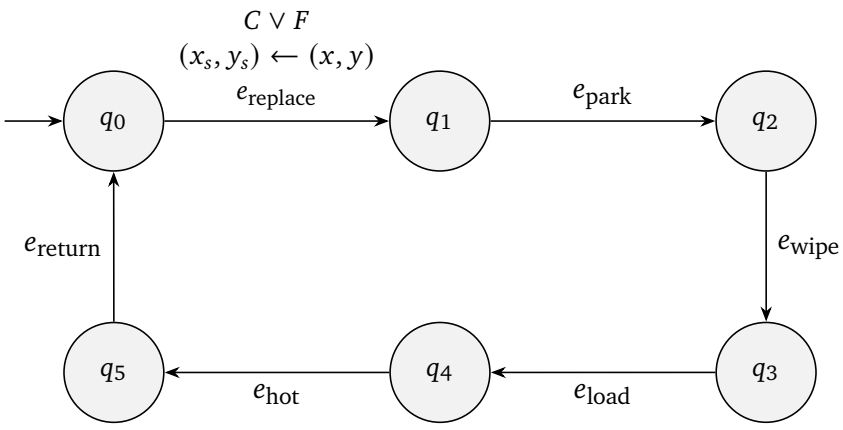


Figure 11: Finite state machine for filament replacement. The initial state is q_0.

Table 2: Controller states.

State	Description
q_0	Printing
q_1	Moving to the wipe tower
q_2	Wiping excess filament
q_3	Loading new filament; heater off
q_4	Heating the nozzle
q_5	Returning to the saved print position

Table 3: Controller events.

Event	Description
e_{replace}	The filament needs to be changed ($C \vee F$), nozzle is turned off, position is saved
e_{park}	The printer head has reached the wipe tower
e_{wipe}	The nozzle has been wiped
e_{load}	New filament is loaded into the nozzle
e_{hot}	Nozzle has reached the desired temperature
e_{return}	Printer head has reached the saved position

State q_0 is selected as the initial state. This state denotes the printing stage and there is no need for a new color to be printed nor a finished spool. When C or F are activated, the current location of the printer head is stored in the variables (x_s, y_s) and the extrusion of the filament is halted. The state becomes q_1 which indicates the process of the printer head moving to the wipe tower. When the printer head reaches the wipe tower event e_{park} has occurred. In state q_2, the excess filament coming from the nozzle is wiped off. The heating of the nozzle is switched off and becomes ready for the loading of new filament as given by q_3. Event e_{load} denotes that new filament is loaded into the nozzle and the nozzle can be heated again. When the nozzle heats up to the required temperature, the system enters state q_5 which returns the nozzle to the print position as saved in (x_s, y_s) and the system has reached the initial state again. There is no accepting state as the 3D-printer continues this cycle until it is turned off.

Question 2 Automata — Parked detection

The automaton for the parked detection is given in Figure 12.

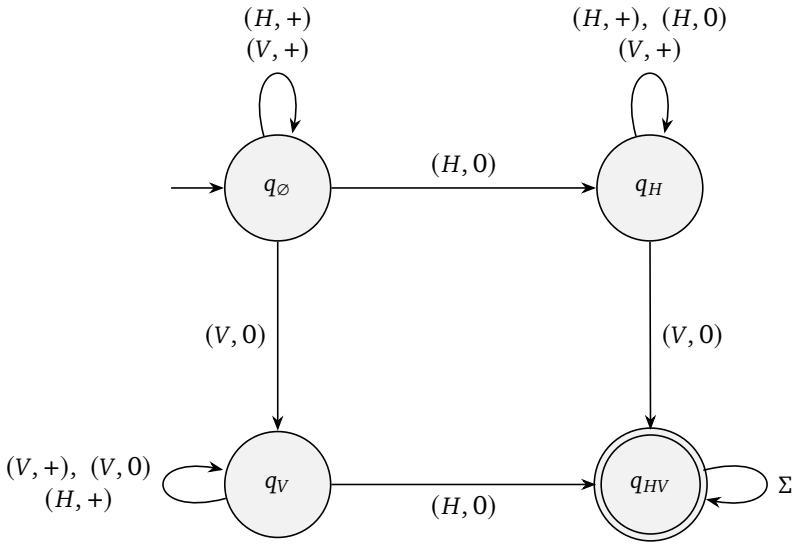


Figure 12: Automaton for detecting the bottom-left parking position. The accepting state q_{HV} indicates that both end positions have been confirmed.

The alphabet corresponding to the automaton is defined as

$$\Sigma = \{(H, +), (H, 0), (V, +), (V, 0)\}. \tag{53}$$

Table 4: Automaton states.

State	Description
$q_{\varnothing}$	Printer head has not reached a boundary
q_H	Printer head has reached the left boundary
q_V	Printer head has reached the bottom boundary
q_{HV}	Printer head has reached the origin.

Table 5: Controller events.

Event	Description
e_{replace}	The filament needs to be changed ($C \vee F$), nozzle is turned off, position is saved
e_{park}	The printer head has reached the wipe tower
e_{wipe}	The nozzle has been wiped
e_{load}	New filament is loaded into the nozzle
e_{hot}	Nozzle has reached the desired temperature
e_{return}	Printer head has reached the saved position

3 Hybrid system

Question 1 **Filament replacement**

Question 2 **Draw a square**

References

[1] Jeremy Lee. “Fourier Pen,” Accessed: Oct. 24, 2026. [Online]. Available: `https://fourier-pen.vercel.app/`

[2] E. den Boer, *Complex fourier series for the efteling logo*, Unpublished CSV file, 2026.

[3] T. Santos et al., “Insights into temperature simulation and validation of fused deposition modeling processes,” *Journal of Manufacturing and Materials Processing*, vol. 7, no. 6, p. 189, 2023.